OCCUPATION SUMS OF THE HARMONIC DESCENT CHAIN AND SLOWLY DECAYING WEIGHTS

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Abstract. Let $a(n,i)$ be the probability that the harmonic descent chain started at n visits i, and $a(i)=\lim_n a(n,i)$. Aldous and Janson asked whether $\sum_{i=2}^{n} a(n,i)f(i)\to\sum_{i\ge 2} a(i)f(i)$ for every non-negative f of polynomial growth with a finite limit sum, and Brandenberger, Chin and Mossel proved this when $f(i)=O(i^{-\varepsilon})$. We show that for every $\delta>0$ there is an f with $f(i)=O((\log i)^{-1+\delta})$ and $\sum_i a(i)f(i)<\infty$ for which the occupation sums are unbounded, along values of n with $f(n)=0$. The cause is that the chain started at n makes, on average, at least of order $\log n/\log\log n$ visits to a window of width $n/(\log n)^2$ just below n.

1. The question

The harmonic descent chain is the Markov chain on $\{1,2,\dots\}$ which, from a state $m\ge 2$, jumps to $m-d$ with probability

$$p(m,m-d)=\frac{1}{d\,h_{m-1}},\qquad 1\le d\le m-1,\qquad h_r=\sum_{j=1}^{r}\frac{1}{j},\tag{1.1}$$

and stays at 1 forever. Started at n, it records the clade sizes on the path from the root to a uniformly chosen leaf of the discrete-time critical beta-splitting tree with n leaves [3, 1, 2]. Let $a(n,i)$ be the probability that the chain started at n ever visits i, so that $a(n,n)=a(n,1)=1$. For fixed $i\ge 2$ the limit

$$a(i)=\lim_{n\to\infty} a(n,i)=\frac{6\,h_{i-1}}{\pi^2(i-1)}\tag{1.2}$$

exists [1, Theorem 18], [2]; Iksanov [5] gave a second proof through the theory of regenerative composition structures, and Chorbadzhiyska, Minchev and Savov [4] express the occupation probabilities of a typical leaf, for the whole beta-splitting family, through the potential measure of a subordinator. Aldous and Janson [1, Open Problem 8] asked for bounds on $|a(n,i)-a(i)|$, and in particular for a proof of the following statement.

Ansatz 20 of [1]**.** For a non-negative sequence $(f(j),\ j\ge 2)$ with $f(j)=O(j^k)$ for some $k<\infty$:

(i) if $\sum_{i\ge 2} a(i)f(i)<\infty$ then $\sum_{i=2}^{n} a(n,i)f(i)\to\sum_{i\ge 2} a(i)f(i)$;
(ii) if $\sum_{i\ge 2} a(i)f(i)=\infty$ then $\sum_{i=2}^{n} a(n,i)f(i)\sim\sum_{i=2}^{n} a(i)f(i)$.

Brandenberger, Chin and Mossel [6] proved (i) for every non-negative f with $f(i)=O(i^{-\varepsilon})$ for some $\varepsilon>0$, as a consequence of their rate $a(n,i)-a(i)=n^{-\beta+o(1)}$; Svihla [7] gives the exact asymptotic expansion of $a(n,i)-a(i)$ for $i=o(\sqrt{n})$. Since $a(n,n)=1$, part (i) as stated fails for any f with $\sum a(i)f(i)<\infty$ and $\limsup f(n)>0$, through the single term $i=n$; Aldous and Janson remark that the method is meant for statistics that are not sensitive to the near-root structure. The point of this note is that the near-root structure matters well beyond that one term: the states just below n are visited many times, and a weight that decays only like a power of $\log i$ can see them.

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Theorem 1.1. *For every $\delta \in (0,1)$ there is a sequence $f \geq 0$ with $f(i) = O((\log i)^{-1+\delta})$ and $\sum_{i\geq 2} a(i)f(i) < \infty$, and integers $N_j \to \infty$ with $f(N_j) = 0$, such that*

$$\sum_{i=2}^{N_j} a(N_j, i) f(i) \longrightarrow \infty.$$

In particular part (i) of Ansatz 20 fails for this f, and it fails for the sum over $2 \leq i \leq n-1$ as well.

The decay $f(i) = O((\log i)^{-1+\delta})$ cannot be replaced by $O(i^{-\varepsilon})$, by [6]. Between the two the question is open; see Remark 3.2.

2. Visits just below the starting point

The chain is strictly decreasing until it reaches 1, so it visits each state at most once and, for a set B of states in $\{2, \dots, n-1\}$,

$$\sum_{i\in B} a(n,i) = \mathbb{E}_n\big[\#\{t \geq 1 :\ X_t \in B\}\big]. \tag{2.1}$$

Lemma 2.1. *Let $n \geq 4$ and let T, L be integers with $1 \leq T \leq L \leq n-2$. Put $x = \lfloor L/T \rfloor$ and $q = h_x/h_{n-1}$. Then*

$$\sum_{i=n-L}^{n-1} a(n,i) \ \geq\ \sum_{t=1}^{T} q^t \ \geq\ T\, q^T.$$

Proof. From any state m with $2 \leq m \leq n$, by (1.1) the jump d satisfies $\mathbb{P}(d \leq x) = h_{\min(x,m-1)}/h_{m-1} \geq h_x/h_{n-1} = q$: if $x \leq m-1$ this holds because $h_{m-1} \leq h_{n-1}$, and if $x \geq m-1$ the probability is 1. Let E_t be the event that each of the first t jumps is at most x. On E_t, for $1 \leq s \leq t \leq T$, we have $n-1 \geq X_s \geq n - sx \geq n - Tx \geq n - L \geq 2$, so the chain has not been absorbed and X_s lies in $B = \{n-L, \dots, n-1\}$. By the Markov property $\mathbb{P}_n(E_t) \geq q^t$. Hence the number of visits to B is at least $\sum_{t=1}^{T} \mathbf{1}_{E_t}$, and (2.1) gives the first inequality. The second holds since $q \leq 1$. □

Corollary 2.2. *Let $N_j = 2^{j^2}$, $L_j = \lfloor N_j/j^4 \rfloor$ and $B_j = \{N_j - L_j, \dots, N_j - 1\}$. Then for all large j,*

$$\sum_{i\in B_j} a(N_j, i) \ \geq\ e^{-9}\Big\lfloor \frac{j^2}{\log j} \Big\rfloor.$$

Proof. Write $\ell = \log N_j = j^2 \log 2$ and take $T = \lfloor j^2/\log j \rfloor$, so $T \leq L_j$ for large j. With $x = \lfloor L_j/T \rfloor$ we have $x + 1 \geq L_j/T \geq N_j/(2j^4 T)$, so, using $h_x \geq \log(x+1)$, $T \leq j^2$ and $h_{N_j-1} \leq 1 + \ell$,

$$q = \frac{h_x}{h_{N_j-1}} \ \geq\ \frac{\ell - 4\log j - \log T - \log 2}{1+\ell} \ \geq\ 1 - u, \qquad u = \frac{2 + 6\log j}{1+\ell}.$$

Then $u \to 0$ and $Tu \leq (2+6\log j)/(\log j \cdot \log 2) \to 6/\log 2 < 8.7$, so $q^T \geq \exp\big(-Tu/(1-u)\big) \geq e^{-9}$ for large j, and Lemma 2.1 gives the bound. □

3. Proof of Theorem 1.1

Fix $\delta \in (0,1)$ and take N_j, L_j, B_j as in Corollary 2.2. Choose j_0 so large that the corollary holds for $j \geq j_0$ and $N_j - L_j > N_{j-1}$ for $j > j_0$; the blocks B_j, $j \geq j_0$, are then disjoint. Set

$$f(i) = j^{-2+2\delta} \quad (i \in B_j,\ j \geq j_0), \qquad f(i) = 0 \text{ otherwise.}$$

Then $f(N_j) = 0$ for every j.

Decay. For $i \in B_j$ we have $\log i \le \log N_j = j^2 \log 2$, so

$$f(i) = (j^2)^{-(1-\delta)} = (\log 2)^{1-\delta} (\log N_j)^{-(1-\delta)} \le (\log i)^{-(1-\delta)}.$$

Finite limit sum. By (1.2) and $h_{i-1} \le 1 + \log i$ there is a constant C with $a(i) \le C \log i / i$ for $i \ge 2$. For $i \in B_j$ we have $i \ge N_j/2$, so

$$\sum_{i \in B_j} a(i) \;\le\; C\,(j^2 \log 2)\,\frac{2\,(L_j+1)}{N_j} \;\le\; \frac{3C}{j^2}$$

for large j, and $\sum_i a(i) f(i) \le \sum_j 3C\, j^{-4+2\delta} < \infty$.

Unbounded occupation sums. By Corollary 2.2,

$$\sum_{i=2}^{N_j} a(N_j, i) f(i) \;\ge\; j^{-2+2\delta} \sum_{i \in B_j} a(N_j, i) \;\ge\; e^{-9}\, j^{-2+2\delta} \Big\lfloor \frac{j^2}{\log j} \Big\rfloor \;\longrightarrow\; \infty.$$

Every term used has $i \le N_j - 1$, which proves the last sentence of the theorem. □

Remark 3.1. The weight in the proof is bounded, so it satisfies the growth condition of Ansatz 20 with $k = 0$. Taking $f = \mathbf{1}_{\bigcup_j B_j}$ (the case $\delta = 1$) gives $\sum_i a(i) f(i) < \infty$ by the same estimate and occupation sums of order at least $j^2/\log j$ at N_j.

Remark 3.2. The proof uses a window of relative width $(\log n)^{-2}$. A block of weight c on a window of relative width w contributes about $c\, w \log n$ to the limit sum, while Lemma 2.1 gives it at least $c \log n / \log\log n$ in the occupation sum when w is at least a power of $(\log n)^{-1}$; for a window $n^{1-\eta}$ the jumps out of the window are no longer rare and the lemma gives only a bounded number of visits. The construction therefore stops short of weights of order $(\log i)^{-1}$, and we do not know whether part (i) holds for every f with $f(i) = O((\log i)^{-1})$, or for every f with $f(i) \to 0$ that is monotone or regularly varying. For monotone f the window just below n carries weight close to $f(n)$, and the question becomes one about the sizes of $a(n, i)$ for i near n, a range not covered by [6] or [7].

4. A computation

By the Markov property, $a(n, n) = 1$ and $a(n, i) = \sum_{m=i+1}^{n} a(n, m) \,/\, \big((m - i)\, h_{m-1}\big)$ for $i < n$; for i in a window $\{n-L, \dots, n-1\}$ only states in the window and n itself contribute, so the window sums can be computed exactly in $O(L^2)$ operations for very large n. Table 1 lists $\sum_{i \in B_j} a(N_j, i)$ for the blocks of Corollary 2.2 and the sums over the top sixteenth $\{n - n/16, \dots, n-1\}$ for $n = 2^{10}, 2^{14}, 2^{18}$. The latter grow by about 0.88 each time n is multiplied by 16, in line with the count of order $\log n$ in Remark 3.2.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [8] and working together inside the Brain framework [10], a shared, persistent

n	window L	$\sum_{i=n-L}^{n-1} a(n,i)$	
2^9	$\lfloor n/3^4 \rfloor = 6$	0.479	$j=3$
2^{16}	$n/4^4 = 256$	1.022	$j=4$
2^{25}	$\lfloor n/5^4 \rfloor = 53{,}687$	1.696	$j=5$
2^{10}	$n/16$	1.407	
2^{14}	$n/16$	2.286	
2^{18}	$n/16$	3.172	

TABLE 1. Expected visits of the harmonic descent chain started at n to the window just below n.

memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [9], the Math Brain repository [8] and the Brain framework repository [10].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the construction and the proofs, wrote and ran the computation of Table 1, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for Table 1, with its output, is available at `https://github.com/yspennstate/harmonic-descent-occupation-sums`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [8].

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